

Visal Chathuranga

AI/ML Engineer · Full-Stack Developer

+94 779 485 361 e19056@eng.pdn.ac.lk LinkedIn GitHub Portfolio

Professional Summary

Aspiring AI Engineer with hands on experience delivering **20+ AI/ML and full-stack projects**, including RAG-based document intelligence platforms, agentic voice AI systems, healthcare EHR solutions, and production grade SaaS applications. B.Sc. in Electrical & Electronic Engineering from the University of Peradeniya *Second Class Upper Division Honours*. Currently developing AI powered healthcare solutions at **MedcubeUSA**, specializing in end to end ML pipelines, LLM integrations, and intelligent automation across FastAPI, Next.js, and cloud-native architectures.

Technical Skills

Languages	Python ▪ TypeScript ▪ JavaScript ▪ HTML ▪ CSS
Frameworks	FastAPI ▪ Next.js ▪ Node.js ▪ Flask ▪ Express.js ▪ React ▪ EJS ▪ Electron.js
AI / ML Libraries	TensorFlow ▪ Keras ▪ PyTorch ▪ Scikit-learn ▪ HuggingFace ▪ LangChain ▪ LangGraph ▪ OpenCV ▪ MediaPipe ▪ Albumentations ▪ timm ▪ LightGBM ▪ NLTK
Data & Viz	Pandas ▪ NumPy ▪ Matplotlib ▪ Seaborn
Web / API Libs	OpenAI SDK ▪ Prisma ▪ Drizzle ORM ▪ jQuery ▪ Tesseract.js
Databases	MongoDB ▪ PostgreSQL ▪ MySQL ▪ Pinecone ▪ Neo4j ▪ ChromaDB ▪ FAIZZ
Cloud & DevOps	AWS (S3 ▪ Textract ▪ Transcribe Medical ▪ EC2) ▪ Vercel ▪ Docker ▪ Kubernetes ▪ Git ▪ CI/CD ▪ DVC ▪ MLflow ▪ RStudio
Integrations	OpenAI API ▪ Claude API ▪ Gemini API ▪ Twilio (Voice & WhatsApp) ▪ Stripe ▪ Playwright ▪ JWT ▪ WebSocket ▪ REST APIs

Experience

Software Engineer (AI/ML) <i>MedcubeUSA</i> Colombo - Architected and deployed enterprise-grade AI solutions including agentic voice assistants, EHR search tools, and intelligent document processing pipelines using RAG, AWS, and Generative AI. - Delivered end to end full-stack SaaS applications from database schema and REST API design to frontend UI covering healthcare workflows, financial automation, and document intelligence platforms. - Owned complete product cycles: requirements gathering, system architecture, AI/ML integration, cloud deployment, and post-launch iteration across FastAPI, Next.js, and cloud-native stacks.	📅 Sep 2025 – Present
AI/ML Engineering Intern <i>GPV Lanka (PVT) Ltd</i> Negombo, Sri Lanka Built AI-powered desktop applications for reflow oven temperature prediction and feeders management using Flask, TensorFlow/Keras, Electron.js, and Python.	📅 Jul 2024 – Oct 2024
Automation Engineering Intern <i>Phoenix Industries Ltd</i> Makandura, Sri Lanka Engineered an automated toothbrush packing machine and a defect detection system for Astra Cups, improving production efficiency and quality control.	📅 Jul 2023 – Oct 2023

Projects

APEX Team: Self Coordinating 14 Agent AI Development System <code></></code> Stack: Python ▪ FastAPI ▪ Next.js 14 ▪ TypeScript ▪ MCP (81 tools) ▪ WebSocket ▪ Playwright ▪ MITRE ATT&CK ▪ NIST CSF 2.0 ▪ Bash / PowerShell - Engineered a self coordinating AI team that spawns 14 specialist agents (Architect, Backend, Frontend, DBA, AI-Integrator, Tester, Security-Auditor, Pen-Tester, DFIR-Analyst, Reviewer, Perf-Tuner, Writer, DevOps, PM) from a single command, sharing a file locked <code>shared_state.json</code> verified at 200 concurrent writes with zero data loss. - Built a unified 81 tool MCP server with a 4 layer intelligence stack token budgeting, shared brain summaries, skill-routing across 2,195+ skills with 17 per role person as and named prompting frameworks achieving 30–70% token reduction via hash based prompt caching. - Deployed a 3 agent security suite (security auditor + pen tester + DFIR analyst) with full finding lifecycle mapped to 6-stage structured debate engine (propose → critique → revise → vote → score → judge), skill-based auto assign, task dependencies, and crash recovery with auto backup. - Shipped a dark themed dashboard with a FastAPI + WebSocket bridge , multi CLI support (Claude Code / Codex / Gemini / Cursor), per agent git worktrees, webhook notifications, and cross platform one-shot installers for Windows, macOS, and Linux.	🔗 GitHub
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VScanMail: AI-Powered Digital Mail Custodian SaaS

🔗 Stack: Next.js ▪ TypeScript ▪ MySQL ▪ Drizzle ORM ▪ OpenAI GPT-4o Vision ▪ AWS Textract ▪ Stripe ▪ JWT ▪ RBAC

- Built a multi-tenant SaaS platform for scanned mail, cheque processing, delivery requests, and deposits with role-based access control.
- Integrated GPT-4o Vision and AWS Textract for extraction, summaries, tamper detection, signature verification, and smart validation.
- Implemented Stripe subscription billing, 2FA secure auth, audit logs, notifications, and vSendDocs delivery API integration.

[🔗 GitHub](#) [🔗 Demo](#)

DocSearch AI: Legal Document Intelligence Platform

🔗 Stack: Node.js ▪ TypeScript ▪ PostgreSQL ▪ MongoDB ▪ Pinecone ▪ OpenAI Embeddings ▪ WebSocket ▪ JWT ▪ Tesseract.js

- Built an AI-powered legal document search and Q&A platform using a RAG pipeline with OpenAI embeddings and Pinecone, enabling semantic search across case laws, labour codes, and regulations.
- Implemented real-time AI chat with WebSocket streaming, OpenAI Assistants API, and per-session vector stores over uploaded legal PDFs with OCR support.
- Full-stack TypeScript/Node.js app with dual databases, PDF processing, admin/client dashboards, and JWT-secured RBAC.

[🔗 Textract Only](#) [🔗 +LLM](#)

Amazon Textract Invoice PDF Processor (Enhanced with LLM)

🔗 Stack: Python ▪ AWS Textract ▪ AWS S3 ▪ Gemini API ▪ JSON

- Built a scalable invoice automation pipeline with S3 integration, MD5 duplicate detection, and intelligent table parsing — achieving **95% accuracy** and reducing manual effort by **80%**.
- Enhanced with a Gemini LLM layer to normalize dates, currencies, and payment terms, improving generalization across complex invoice formats.

[🔗 GitHub](#)

MedScribe Pro: Medical Transcription Platform

🔗 Stack: FastAPI ▪ AWS Transcribe Medical ▪ AWS S3 ▪ OpenAI GPT-4 ▪ Python

- Built a FastAPI platform converting medical audio (.mp3/.webm) into role-labeled transcripts (Doctor/Patient/Nurse) using AWS Transcribe Medical and GPT-4 for clinical summaries.
- Optimized with S3 Transfer Acceleration, multipart uploads, concurrent workers, and speaker diarization for real-time performance.

[🔗 GitHub](#)

WhatsApp Petty Cash Management System

🔗 Stack: FastAPI ▪ Next.js ▪ Twilio WhatsApp API ▪ AWS Textract ▪ OpenAI GPT-4 ▪ Docker ▪ JWT ▪ RBAC

- Built a full-stack expense system with a conversational WhatsApp bot, FastAPI MVC backend, and Next.js admin panel featuring AI-powered OCR and a dynamic rules engine.
- Architected multi-tenant SaaS with RBAC, organization provisioning, audit logging, and dual channels (WhatsApp AI + web) for state-managed claim workflows.
- Deployed Dockerized production system with OCR-validated claims, difference warnings, and a comprehensive admin approval interface.

[🔗 GitHub](#) [🔗 Demo](#)

EHR AI Search: Healthcare Intelligence Platform

🔗 Stack: FastAPI ▪ React ▪ OpenAI GPT-4o Mini ▪ WebSocket ▪ Vercel ▪ Playwright ▪ JWT

- Built an AI-powered EHR search platform using GPT-4o Mini with natural language queries, React chat widget, and voice I/O support.
- Deployed as an embeddable Vercel widget with FastAPI backend, WebSocket streaming, and Playwright-based auto-token refresh.

[🔗 GitHub](#)

AI Voice Agent with RAG & Telephony

🔗 Stack: FastAPI ▪ OpenAI Realtime API ▪ Twilio Voice ▪ Pinecone ▪ WebSocket ▪ Python

- Built a production-ready voice AI system integrating OpenAI Realtime API with Twilio telephony for natural phone conversations with real-time audio streaming.
- Implemented a Pinecone-based RAG pipeline with dual modes: Full Agentic (RAG + web search + tools) and RAG-Only for compliance-critical industries.

Diabetic Retinopathy Detection: Deep Learning Classifier

🔗 Stack: Python ▪ PyTorch ▪ EfficientNetB3 ▪ timm ▪ Albumentations ▪ Scikit-learn

- Built an automated 5-class DR severity classifier using EfficientNetB3 with Ben Graham preprocessing, achieving **QWK 0.83** on APTOS 2019 and 0.57 on EyePACS (35K+ images).
- Engineered two-phase training pipeline with mixed precision (fp16), WeightedRandomSampler, patient-level splitting, and Test Time Augmentation.
- Designed a session-resilient training system with auto-resumable checkpointing to Google Drive for free-tier GPU environments (Colab T4/Kaggle P100).

[🔗 GitHub](#) [🔗 Demo](#)

YouTube Sentiment Insights

🔗 Stack: Python ▪ Flask ▪ LightGBM ▪ DVC ▪ MLflow ▪ Docker ▪ AWS EC2 ▪ GitHub Actions

- Developed an AI-powered Chrome Extension and Flask backend for real-time YouTube comment sentiment analysis, trends, and

dashboards.

- Built a scalable ML pipeline with DVC, MLflow, and LightGBM; Dockerized backend deployed on AWS EC2 with GitHub Actions CI/CD.

 [GitHub](#)

Industrial AI: Reflow Oven Profile Predictor

 **Stack:** Electron.js ■ Python ■ Flask ■ TensorFlow ■ Keras

- Designed a hybrid desktop application with Electron.js frontend and Python/Flask backend, integrating TensorFlow/Keras neural networks to optimize industrial reflow oven soldering profiles.

Education

B.Sc. Engineering (Hons.) — Electrical & Electronic Engineering

2021 – 2025

GPA: 3.4 / 4.0

University of Peradeniya

Second Class Upper Division Honours

GCE Advanced Level

2010 – 2018

Dharmasoka College, Ambalangoda

Mathematics: A ■ Chemistry: A ■ Physics: B

Certifications

- Generative AI Mastery with 15+ Real Time Projects — [Udemy](#)
- Deep Learning Neural Networks & Image Processing — [edXcope](#)
- Machine Learning with Python — *with Honors* — [Coursera](#)
- IBM AI Engineering Specialization — [IBM](#) / [Coursera](#)
- Deep Neural Networks with PyTorch — *with Honors* — [IBM](#) / [Coursera](#)
- Building Deep Learning Models with TensorFlow — [IBM](#) / [Coursera](#)
- AI Capstone Project with Deep Learning — [IBM](#) / [Coursera](#)
- Introduction to Deep Learning & Neural Networks — [IBM](#) / [Coursera](#)
- Introduction to Computer Vision & Image Processing — [IBM](#) / [Coursera](#)
- Deep Learning and Reinforcement Learning — [IBM](#) / [Coursera](#)
- NLP with Classification and Vector Spaces — [DeepLearning.AI](#)
- NLP with Probabilistic Models — [DeepLearning.AI](#)
- Introduction to Web Development (HTML/CSS/JS) — [IBM](#) / [Coursera](#)
- Complete Java Full Stack Training — [Digimax Education Sri Lanka](#)

Extracurricular Activities

Treasurer	IEEE Microwave Theory and Techniques Society (MTTS)	Jan 2024 – Aug 2025
Team Member	IEEE MTTS — Programming Team	Jan 2023 – Jan 2024
Member	Explorer's Club, University of Peradeniya	2022 – Present
Volunteer	Arunella Social Development Initiative	2022 – 2025

Supporting school students in underprivileged areas.

Soft Skills & Interests

Soft Skills Adaptability ■ Problem-Solving ■ Teamwork ■ Communication ■ Creativity ■ Time Management

Interests Exploring New Technologies ■ Playing Chess ■ Environmental Exploration ■ Watching Films

Non-Related Referees

Dr. Tharindu Weerakoon

Senior Lecturer

Dept. of Electrical & Electronic Engineering

University of Peradeniya, Sri Lanka

✉ tharinduw@eng.pdn.ac.lk

Dr. Ruwan Ranaweera

Senior Lecturer

Dept. of Electrical & Electronic Engineering

University of Peradeniya, Sri Lanka

✉ rdbanaweera@ee.pdn.ac.lk